DAS VALIDATION MOP

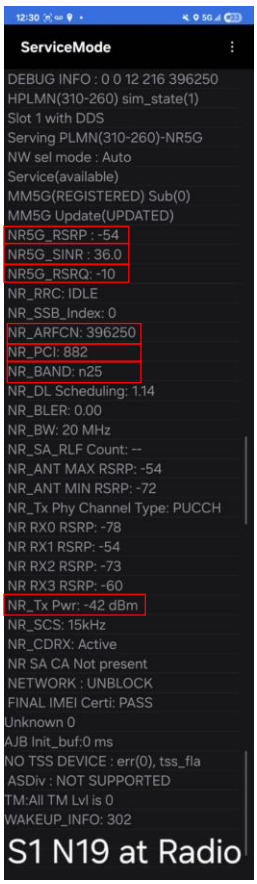
For FEs:

- **SERVICE MODE TESTS:**

Passing Thresholds -

Tests	RSRP	SINR	RSRQ	TX Power
Radio End (dummy antenna connected to the Radio port)	-60 dBm to -40 dBm	Above 25	-12 to -3	Always Negative
Antenna End (under DAS antennas)	-75 dBm to -50 dBm and above	Above 25	-12 to -3	Always Negative

Sample Screenshot:



- **SPEED TESTS:**
  Perform Speed Tests on individual band along with LTE All, NR All/Float (Selecting All Bands)

Passing Thresholds as per BW/Layers –

	BANDWIDTHS			DL RANGE				UL RANGE			PROGRESSIVE UL BASED ON SINR LEVEL
LAYERS	BW LTE	BW NR C1	BW NR C2	LTE DL	NR DL	EN-DC DL	NR-DC DL	LTE UL	NR UL	EN-DC UL	EN-DC UL SINR (15/18/20/22/24/26/28/30)
SISO	5	5	0	11-20	10-19	21-39	21-32	2-5	1-4	4-10	4/4/5/6/7/8/9/10
SISO	10	10	0	22-41	22-41	43-82	46-70	5-11	3-9	8-21	8/9/10/12/14/17/19/21
SISO	15	15	0	32-61	34-65	67-126	73-111	8-17	5-15	12-32	12/14/16/19/22/26/29/32
SISO	20	20	0	43-82	46-87	89-169	98-149	10-23	6-20	16-43	16/18/22/25/29/35/39/43
SISO	20	30	0	43-82	70-134	113-215	128-207	10-23	10-30	20-54	20/22/27/31/36/43/48/54
SISO	20	40	0	43-82	96-182	139-263	160-267	10-23	13-41	23-65	23/27/33/38/43/52/57/65
SISO	20	50	0	43-82	120-228	163-309	190-325	10-23	16-52	27-75	27/31/38/44/50/60/66/75
SISO	20	60	0	43-82	146-277	189-359	223-387	10-23	20-63	30-87	30/36/44/50/57/69/76/87
SISO	20	70	0	43-82	170-324	214-405	253-445	10-23	23-74	33-97	33/40/49/56/64/77/85/97
SISO	20	80	0	43-82	196-372	239-453	285-505	10-23	27-85	37-108	37/44/55/62/71/86/94/108
SISO	20	90	0	43-82	221-419	264-501	317-565	10-23	30-96	40-119	40/49/61/69/78/95/104/119
SISO	20	100	0	43-82	246-467	289-549	348-624	10-23	34-107	44-130	44/53/66/75/86/103/113/130
SISO	20	100	40	43-82	342-659	385-731	468-851	10-23	47-148	57-171	57/70/87/99/112/136/148/171
SISO	20	100	60	43-82	392-745	435-827	531-971	10-23	54-170	64-193	64/79/99/111/127/153/167/193
SISO	20	100	80	43-82	442-839	485-921	593-1089	10-23	60-191	70-215	70/87/110/123/140/170/186/215
SISO	20	100	90	43-82	688-1306	731-1388	624-1149	10-23	64-202	74-226	74/87/110/123/140/170/186/226

	BANDWIDTHS			DL RANGE				UL RANGE			PROGRESSIVE UL BASED ON SINR LEVEL
LAYERS	BW LTE	BW NR C1	BW NR C2	LTE DL	NR DL	EN-DC DL	NR-DC DL	LTE UL	NR UL	EN-DC UL	EN-DC UL SINR (15/18/20/22/24/26/28/30)
MIMO	5	5	0	21-39	20-38	40-77	42-64	2-5	1-4	4-10	4/4/5/6/7/8/9/10
MIMO	10	10	0	41-78	43-82	85-160	92-141	5-11	3-9	8-21	8/9/10/12/14/17/19/21
MIMO	15	15	0	62-117	69-130	130-247	146-223	8-17	5-15	12-32	12/14/16/19/22/26/29/32
MIMO	20	20	0	82-156	92-175	174-331	195-299	10-23	6-20	16-43	16/18/22/25/29/35/39/43
MIMO	20	30	0	82-156	141-267	223-424	256-414	10-23	10-30	20-54	20/22/27/31/36/43/48/54
MIMO	20	40	0	82-156	191-363	274-519	320-534	10-23	13-41	23-65	23/27/33/38/43/52/57/65
MIMO	20	50	0	82-156	240-455	322-612	380-650	10-23	16-52	27-75	27/31/38/44/50/60/66/75
MIMO	20	60	0	82-156	292-555	375-711	446-774	10-23	20-63	30-87	30/36/44/50/57/69/76/87
MIMO	20	70	0	82-156	341-647	423-804	507-889	10-23	23-74	33-97	33/40/49/56/64/77/85/97
MIMO	20	80	0	82-156	392-743	474-899	570-1009	10-23	27-85	37-108	37/44/55/62/71/86/94/108
MIMO	20	90	0	82-156	442-839	524-995	633-1129	10-23	30-96	40-119	40/49/61/69/78/95/104/119
MIMO	20	100	0	82-156	493-935	575-1091	696-1249	10-23	34-107	44-130	44/53/66/75/86/103/113/130
MIMO	20	100	40	82-156	684-1298	766-1454	935-1703	10-23	47-148	57-171	57/70/87/99/112/136/148/171
MIMO	20	100	60	82-156	785-1490	867-1646	1062-1942	10-23	54-170	64-193	64/79/99/111/127/153/167/193
MIMO	20	100	80	82-156	884-1678	966-1834	1186-2178	10-23	60-191	70-215	70/87/110/123/140/170/186/215
MIMO	20	100	90	82-156	935-1774	1017-1930	1249-2298	10-23	64-202	74-226	74/87/110/123/140/170/186/226

Sample Screenshots:

Best Practices –

- To obtain a negative TX power reading, initiate a call (either to E911 or to your personal number) and capture a screenshot of the service mode display showing the negative TX power value.
- For NR testing at the radio end with dummy antenna, if RSRP or SINR readings are not observed, increase your distance from the dummy antenna before retesting.
- If the Radio end test values are not within the specified range, inspect the dummy antenna connection and re-torque as necessary.
- Confirm with back-office support whether the system is configured as SISO or MIMO and verify the bandwidth for each band before testing speeds.
- If the values under the DAS antennas are not within the specified range, relocate to an antenna positioned at the center of the building, away from nearby windows and free from macro interference. Conduct tests under different antennas and on multiple floors for comparison.
- Always double-check all parameter readings (RSRP, RSRQ, SINR, PCI, UE TX Power) along with DL/UL speeds before capturing screenshots.
- Label each screenshot with Sector ID, Technology, and location of the test - for example, 'at Radio' (if performed at the radio end) or 'TP1/TP2' (Test Point 1/2 if performed under the antenna).
- Please ensure that the timestamps between the service mode and speed test results are within a maximum interval of 3-4 minutes.
- Finally, having a Positive approach toward completing the target scope of work for the day!

For IX/Back Office Team:

- External alarms must be configured for each site.
- VoNR CBN logs must be provided of Live/PSAP NR calls.
- For DRAN sites, once IX is completed, conduct testing with AILG ON for a duration of 2 hours during the maintenance window.